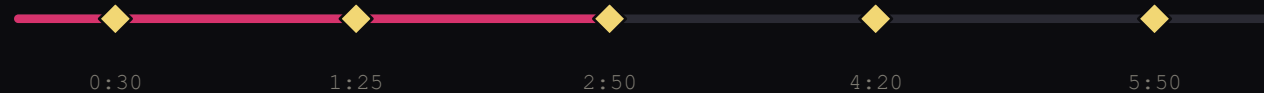


LFC 2026 · AI UNLEASHED · FUTURE LEARNING CONCEPT · GROUP 12

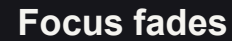
# COGNIVUE

*See what you missed. Master what matters.*

AI concept checkpoints inside Echo360 lecture recordings.



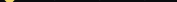
***“Students need a way to check their understanding while watching the lecture, without leaving the platform.”***



## Hard to find again

## Revision becomes rewatching

## AI tools sit outside



```
graph LR; Step1[1 AI maps the lecture] --> Step2[2 The lecture pauses to check]; Step2 --> Step3[3 Every answer goes somewhere];
```

**1 AI maps the lecture**  
Transcript and slides scanned; every concept pinned to its timestamp.

**2 The lecture pauses to check**  
A 30-second diagnostic quiz, then straight back to the video.

**3 Every answer goes somewhere**  
Correct continues. Incorrect opens an explanation, resources and a one-click jump back.

| Time | Topic                      |
|------|----------------------------|
| 0:30 | Independence of motion     |
| 1:25 | Horizontal motion          |
| 2:50 | Vertical motion & the peak |
| 4:20 | Time of flight & range     |
| 5:50 | Real-world application     |

# A wrong answer becomes a thirty-second lesson

◆ CHECKPOINT 3 · 2:50

## Vertical motion & the peak

A ball is thrown straight up at 30 m/s. Taking  $g = 10 \text{ m/s}^2$ , how long does it take to reach the peak?

A. 0.33 s *the classic slip: dividing the wrong way ( $g \div v_0$ )*



B. 3 s *correct: at the peak,  $v = 0$ , so  $t = v_0 / g = 30 / 10$*



C. -3 s *a sign error in  $v = v_0 - gt$*



### TAKE NOTE

At the peak, vertical velocity is zero:  $t = v_0 / g = 3 \text{ s}$ . Saved to your notes · pinned to 2:50

▶ REPLAY 2:50

DIAGRAM

WORKED EXAMPLE

SIMPLIFIED



### Instant feedback

Correct continues the lecture. Wrong opens the answer and a plain-English why.



### Targeted help, not a rewatch

One click replays 2:50, with linked diagrams and worked examples.



### Take Note, pinned forever

A one-line summary, saved to the exact timestamp.

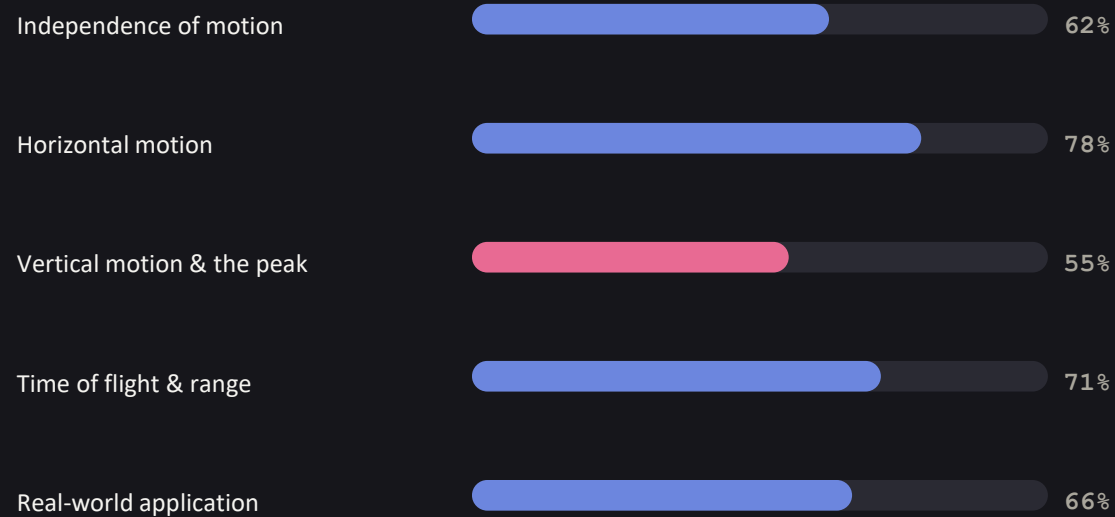


### Retry, skip, or switch off

Retries earn half points. Watch-only mode never interrupts.

## Exactly what to revise: you against the class

## Concept results · Week 1 · Projectile Motion



First-attempt accuracy · weakest concept highlighted



## Focus next

Weak topics flagged automatically, with rewatch links attached.



## Open to review

Skipped or missed checkpoints queue for retry.



## Your checkpoint notes

Every Take Note card, pinned to its timestamp. One click rewatches.



# The same lecture, a different pathway for every student



## Personalised learning

Every student's pathway shifts with their understanding.



## Intelligent feedback

Short diagnostic questions after each concept, marked in real time.



## Adaptive teaching

Every gap gets a personalised explanation, not a generic one.



## Real-world applications

Concepts grounded in relatable scenarios, tailored to the degree.

### Every answer changes the path

✓ **Correct** continues the lecture, no interruption.

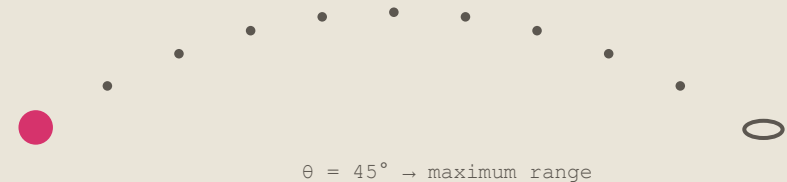
✗ **Incorrect** opens adaptive support:

[REPLAY](#)[DIAGRAM](#)[WORKED EXAMPLE](#)[SIMPLIFIED](#)

*A re-check follows, and the pathway adapts until it sticks.*

### REAL-WORLD APPLICATION · GENERATED FOR 2:50

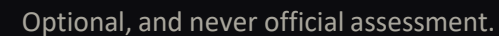
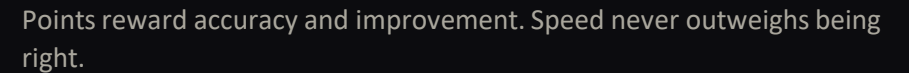
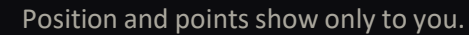
*"Think of a basketball shot. The launch angle decides the arc: past 45° you gain height but lose range."*



2,700 pts available · opt-in · never official marks

*Rank and points stay private.*

Everyone else shows as “Anonymous student”. Names are opt-in.



## See what the cohort struggled with, before the exam

87%

of 148 students used checkpoints,  
entirely optional

66%

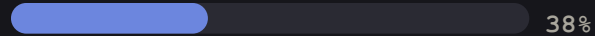
average first-try accuracy, Week 1

2

misconceptions diagnosed from Week 1  
wrong answers

**Missed on first attempt · Week 1**

## Independence of motion



## Horizontal motion



## Vertical motion & the peak



## Time of flight & range



## Real-world application



Share answering incorrectly · anonymised · higher = revisit

### Most common wrong answers

29%

## Time of flight & range · 4:20

Used  $g = 9.8$  after being told  $g = 10$ , or a sign error.

22%

## Horizontal motion · 1:25

“Gravity plus the throw force”: the impetus misconception.



**A weekly digest, written for the lecturer**

One email: what was missed, why, and what to revisit in lectures, tutorials or revision materials. Aggregated and anonymised.



## Revision arrives when it matters: Week 5 and Week 12

From Cognivue <checkpoints@cognivue.app>  
To you <student@curtin.edu.au>  
Subject **Week 5 progress: PHYS1001, ready for your mid-semester test**

## Your checkpoint performance so far

- Week 1 (Projectile Motion): 5 / 5 checkpoints · first-try 80%
- Week 2 (Forces & Newton's Laws): 4 / 5 checkpoints · first-try 75%
- Total points: 1,120 · best streak: 6

## Topics to revise before your mid-semester test

- Vertical motion & the peak: missed first try. Jump to 2:50 →
- Newton's third law: skipped. Retry the checkpoint at 4:05 →

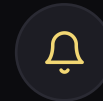
## Suggested plan

- Rewatch the two explanations above (one click each)
- Retry wrong or skipped checkpoints (half points, streak rebuilds)
- Try the challenge questions for exam-style practice



## Week 5 · mid-semester

Strengths, gaps and a plan, before the mid-semester test.



## Week 12 · pre-exam

Every weak concept ranked, with direct timestamp links.



## No rewatching marathons

Every line links to a moment. Minutes of recall, not hours of video.



# Built where students already watch: inside the Echo360 player

echovideo

PHYS1001 Mechanics · Semester 2 2026 · Bentley · Week 1

◆ COGNIVUE · AI ADD-ON

◆ 0 pts

PHYS1001 Mechanics (2026 S2) 105.413:LT · Week 1: Projectile Motion · Dr Elena Marsh

PHYS1001 MECHANICS · WEEK 1

◆ AI · PAUSED

# Projectile Motion

Dr Elena Marsh · Recorded lecture · Semester 2

- ◆ Horizontal and vertical motion, step by step
- ◆ Time of flight, range, and the 45° result

010

0:00 / 7:30

◆ NEXT CHECKPOINT

FINISH & VIEW SUMMARY ►►

CC

4x

LECTURE MODE

✓ Interactive

Watch only

Interactive pauses at key concepts · Watch only never interrupts

## Week 1: Projectile Motion

AI has scanned this recording's transcript & slides and placed **5** concept checkpoints.

POINTS 0    ACCURACY N/A    STREAK 0

### Concept checkpoints

|      |                            |           |
|------|----------------------------|-----------|
| 0:30 | Independence of motion     | Take quiz |
| 1:25 | Horizontal motion          | Take quiz |
| 2:50 | Vertical motion & the peak | Take quiz |
| 4:20 | Time of flight & range     | Take quiz |
| 5:50 | Real-world application     | Take quiz |

Recreated from the working demo built this challenge week.

▶ [OPEN THE LIVE DEMO](#)

COGNIVUE · GROUP 12 · FUTURE LEARNING CONCEPT

◆ ◆ ◆ ◆ ◆ ◆ ◆ ◆ ◆ ◆

## STUDENTS

Revision in minutes, not hours of rewatching.

## LECTURERS

Misconceptions surface weekly, in time to reteach before the exam.

## THE UNIVERSITY

More learning from the lecture platform Curtin already runs.

# COGNIVUE

*See what you missed. Master what matters.*



All checkpoints complete

▶ [OPEN THE LIVE DEMO](#)

[anishboodia-dot.github.io/ai-ilecture-checkpoints-demo](https://anishboodia-dot.github.io/ai-ilecture-checkpoints-demo)

GROUP 12 · FUTURE LEARNING CONCEPT · LFC 2026 · AI UNLEASHED · CURTIN UNIVERSITY

*Concept prototype with demo content only. Not affiliated with Echo360 or Curtin University.*